

Total No. of Questions : 8]

SEAT No. :

PC-2635

[Total No. of Pages : 2

[6354]-784

B.E. (Artificial Intelligence and Data Science)
INDUSTRIAL INTERNET OF THINGS
(2019 Pattern) (Semester - VII) (417523B) (Elective - III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8
- 2) Neat diagrams must be drawn wherever necessary
- 3) Figures to the right indicate full marks
- 4) Assume suitable data if necessary.

- Q1) a)** What are the primary components of an industrial IoT system? Explain each in details. [6]
- b) Explain the Industrial Internet Architecture Framework (IIAF). [6]
- c) What are different types Industrial IoT Layers? Explain in details each layer. [6]

OR

- Q2) a)** What are the key components and Design consideration of a typical WSN used in industrial IoT? [6]
- b) Explain the Business Model Architecture of industrial IoT. [6]
- c) Explain the difference between edge computing and cloud computing in industrial IoT. [6]

- Q3) a)** Explain the concept of Cloud of Things (CoT) and its significance in the context of Industrial IoT. [6]
- b) What is the role of data analytics in Industrial IoT? Explain Any one Example. [6]
- c) What is need of Digital Twin for industrial IoT? Explain Elements of Digital Twin in Industrial IoT. [6]

OR

P.T.O.

- Q4)** a) Explain the SaaS PaaS and IaaS with examples. [6]
b) Explain the importance of data visualization techniques in presenting complex Industrial IoT data. [6]
b) Provide examples of how digital twin technology is transforming industries and driving digital transformation. [6]

- Q5)** a) What are the key challenges and security considerations when implementing the industrial IoT in industrial environments? [6]
b) What is the concept of threat analysis in the context of Industrial IoT? Explain the industrial IoT threat analysis in smart manufacturing plant. [6]
c) Explain the Network security techniques in industrial IoT. [5]

OR

- Q6)** a) Explain the Conventional web technology and relationship with industrial IoT. [6]
b) What is importance of access control in industrial IoT? Explain the methodologies for access control. [6]
c) Explain Management aspects of cyber security in industrial IoT. [5]

- Q7)** a) Explain the concept of smart factories and how they differ from traditional manufacturing facilities. [6]
b) How does industry 4.0 benefits the industrial IoT in manufacturing process? [6]
c) What is industry 5.0 with example? [5]

OR

- Q8)** a) Explain the health care application in industrial IoT. [6]
b) Describe the advantages and challenges of implementing smart logistics in supply chain management. [6]
c) What are Key characteristics of industry 4.0? [5]

